Unit 55 Quantitative methods of analysis

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

55.1 Quantitative analysis

The discipline of analytical chemistry consists of qualitative and quantitative analyses. In Units 53 and 54, we have looked at qualitative analysis. Here we are going to look at quantitative analysis, i.e. determining how much a constituent is present in a sample. Two common methods used are **gravimetric analysis** and **volumetric analysis**.

Extension

Gravimetric analysis

Gravimetric analysis involves determining the mass of the species being analyzed or the mass of a compound chemically related to that species. The precipitation method in gravimetric analysis involves precipitating the species to be analyzed by a reagent.

For example, we can analyze the concentration of chloride ions in a sample solution by precipitating the chloride ions with an excess aqueous solution of silver nitrate.

Chloride ions react with the silver ions according to the following equation:

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

According to the equation, 1 mole of chloride ions reacts with 1 mole of silver ions to produce 1 mole of silver chloride.

As excess silver nitrate is used, all chloride ions in the sample solution should be precipitated. Then we can calculate the concentration of chloride ions in the sample solution from the mass of the precipitate.

Volumetric analysis

Volumetric analysis involves measuring the volume of a solution containing sufficient reagent to react completely with the species being analyzed in a process called **titration**. N1

Extension

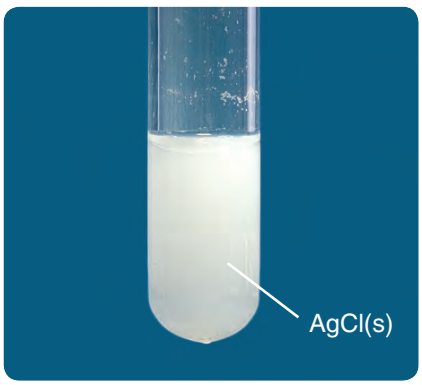


Fig. 55.1 Chloride ions in a sample solution can be precipitated as AgCl(s) by excess aqueous solution of silver nitrate

N1 Reaction that can be used in volumetric analysis

A reaction must fulfill certain conditions so that it can be used in volumetric analysis, including:

- a significant observable change can easily be detected upon the complete consumption of the species being analyzed, i.e. a suitable indicator is available;
- the reaction must be near to completion;
- the reaction must be fast.

gravimetric analysis 重量分析　volumetric analysis 容量分析　titration 滴定

During the titration, we gradually add a solution of known concentration of the reagent (a **titrant**) to the species being analyzed until the species reacts completely. We can use the volume of added titrant to calculate the amount or concentration of the species being analyzed present in the sample.

For example, the following analyses involve acid-alkali titrations*:

- determining the ammonia content in glass cleanser;
- determining the ethanoic acid content in vinegar;
- determining the citric acid content in orange juice.

We can also apply a similar titration procedure to redox reactions — a process called redox titrations*.

Choosing an analytical method

The first step in an analysis is to pick a suitable method of analysis. We use a qualitative method if the aim is simply to find out the species present in a sample. However, if the aim is to find out how much each species is present, then a quantitative approach is essential.

For a quantitative analysis, we need to select a method based on factors such as:

- the sample type;
- the sample size available;
- concentration range of the species being analyzed — we may use a more sensitive* method when determining trace components in a sample;
- the accuracy* required;
- separation that may be required to eliminate interferences — for example, eliminating impurities may be important when determining trace components in a sample;
- the instruments available;
- the cost;
- the speed.

* We have learnt about acid-alkali titrations in Topic 4 Acids and Bases.

* We will discuss redox titrations later in this unit.

* The **sensitivity** of a method is the ability to distinguish between two different concentrations.

* The **accuracy** of a method is the closeness of the measured value to the true value for the sample.

titrant 滴定劑　　sensitivity 敏感度　　accuracy 準確度



Fig. 55.2 An oven cleanser

When deciding on an analytical method, we need to consider the properties of the species being analyzed as well. For example, the analysis of an oven cleanser containing the alkali sodium hydroxide is likely based on the reaction of the alkali with an acid. On the other hand, the analysis of a household bleach containing the oxidizing agent sodium hypochlorite is likely based on the reaction with a suitable reducing agent.

As in qualitative analysis, it is important to carry out a risk assessment when we plan to use any one of the analytical methods.

Extension

55.2 Steps of the precipitation method in gravimetric analysis

We can summarize the steps of the precipitation method in gravimetric analysis as follows:

1 accurately weigh the sample*;

2 dissolve the sample*;

3 precipitate the species being analyzed by adding a suitable precipitating reagent;

4 allow the precipitate to stand in the hot **mother liquor** (i.e. the solution from which it is precipitated) for a period of time*;

5 filter off the precipitate;

6 wash the precipitate to remove the impurities;

7 dry the precipitate;

8 weigh the precipitate;

9 calculate the percentage by mass of the species being analyzed from masses of the sample and the precipitate.

* We can skip these two steps if the sample is a liquid already.

* The small particles tend to dissolve and reprecipitate on the surfaces of larger particles. Particle size of the precipitate increases and impurities tend to be expelled.

CE 2002 Paper 1 Q7(a)(iv)
CE 2007 Paper 1 Q5(c)

Filtering off the precipitate

In gravimetric analysis, it is necessary to collect all of the precipitate. We may collect the precipitate in a **sintered glass filtering crucible** (Fig. 55.3). We can complete the filtration, drying and weighing steps in this single vessel. The porous glass bottom of the crucible allows liquid to pass but retains solids.

mother liquor 母液　　sintered glass filtering crucible 燒結玻璃過濾器

Extension

Drying the precipitate

After filtration, the precipitate is heated to dry until its mass becomes constant. Heating removes the solvent and any volatile impurities carried down with the precipitate. This is often carried out in an oven*.

To minimize the uptake of moisture, store dried materials in **desiccators** while they cool.

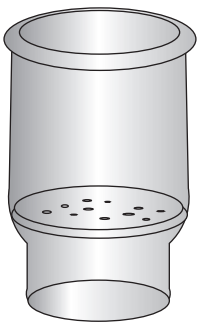


Fig. 55.3 A sintered glass filtering crucible



Fig. 55.4 A desiccator

Calculating the percentage by mass of the species being analyzed in sample

In gravimetric analysis, we are generally interested in the percentage by mass of the species being analyzed in the sample.

$$\text{Percentage by mass of species being analyzed in sample} = \frac{\text{mass of species being analyzed (g)}}{\text{mass of sample (g)}} \times 100\%$$

55.3 Possible major sources of error

Every measurement has some uncertainty, which is called experimental error. We can classify an experimental error as either systematic or random*.

* Heating in the oven is not suitable for substances that decompose at the temperature of the oven.

* We have discussed this in Topic 13 Investigative Study in Chemistry.

desiccator 乾燥器

Extension

Systematic error

Systematic error arises from a flaw in equipment, operations during the experiment or the design of the experiment. We cannot eliminate such an error by repeating the measurement. In general, a systematic error in a series of repeated measurements causes all the results to be always above or always below the true value.

There are three common types of systematic error: N2

- instrumental error;
- operative error;
- error of method*.

Here are some possible ways to minimize the error in gravimetric analysis:

- Minimize the error due to the loss of precipitate by
  - — producing a precipitate of sufficiently low solubility;
  - — using a minimum amount of solvent to wash the precipitate.
- Minimize the error due to incomplete drying of the precipitate by a procedure called weighing to a constant mass — this is to weigh the dried precipitate, then dry and weigh it again, until successive readings are the same.

Random error

Random error causes measurements to be distributed randomly on either side of the true value. Random error exists in every measurement.

We can never eliminate the effect of random error totally. However, we can reduce the effect by repeating the measurements and taking a mean.

* This is the most serious error of an analysis. When the error becomes intolerable, we must use another method for the analysis.

N2 Common types of systematic error:
- Instrumental error — All measuring devices are potential sources of systematic error. For example, pipettes and burettes may hold or deliver volumes slightly different from the volumes indicated by the manufacturer.
- Operative error — This includes a personal error. Many measurements require personal judgments. For example, during a titration, a chemist who is insensitive to colour changes tends to use excess reagent.
- Error of the method — Such errors include the slight solubility of a precipitate, the slowness of reactions, incomplete reactions and impurities in the reagents.

systematic error 系統誤差　　random error 隨機誤差

Extension

55.4 Gravimetric determination of phosphorus content in a sample of fertilizer

* This way of showing the amount of nutrients was used in early laws and it has remained in use even when fertilizers are not made of oxides.

Plants, like animals, require essential nutrients to live. If the soil lacks these nutrients, we can supply them through the use of fertilizers. For example, a fertilizer labelled 10.25.15* contains

- 10.0% by mass of nitrogen;
- 25.0% by mass of phosphorus pentoxide (P_2O_5); and
- 15.0% by mass of potassium oxide (K_2O).

10.25.15	
NPK fertilizer	10.25.15
Nitrogen (N)	10.0%
Phosphorus pentoxide (P_2O_5)	25.0% (10.9% P)
Potassium oxide (K_2O)	15.0% (12.4% K)



Fig. 55.5 A fertilizer labelled 10.25.15



Fig. 55.6 The label on a box of fertilizer

Extension

* If an acid is added to an aqueous solution containing PO_4^{3-} ions, the following three processes occur, each at a lower pH than the preceding one:

$PO_4^{3-}(aq) + H^+(aq) \rightleftharpoons HPO_4^{2-}(aq)$

$HPO_4^{2-}(aq) + H^+(aq) \rightleftharpoons H_2PO_4^-(aq)$

$H_2PO_4^-(aq) + H^+(aq) \rightleftharpoons H_3PO_4(aq)$

* If the solution is too basic, Mg^{2+} ions will precipitate as $Mg(OH)_2$.

The phosphorus exists as phosphate ions (PO_4^{3-}) in the fertilizer. When the fertilizer is dissolved in water, the solution may contain the following species: PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$ or H_3PO_4*.

We can determine the phosphorus content in a sample of fertilizer by precipitating the phosphorus as magnesium ammonium phosphate ($MgNH_4PO_4 \bullet 6H_2O$). The precipitate is produced in a solution containing monohydrogen phosphate ions (HPO_4^{2-}), ammonium ions and magnesium ions.

$$Mg^{2+}(aq) + NH_4^+(aq) + HPO_4^{2-}(aq) + OH^-(aq) + 5H_2O(l) \longrightarrow MgNH_4PO_4 \bullet 6H_2O(s)$$

The difficult part of this experiment is to control the acidity of the solution so that the only phosphate species present is the HPO_4^{2-} ions.

If the solution is too basic, PO_4^{3-} ions rather than HPO_4^{2-} ions will be present and no precipitate will form*. Therefore during the experiment, we need to add aqueous ammonia slowly until the solution just begins to become basic.

The experimental details are as follows:

Step 1 Preparing the sample solution	Accurately weigh about 3 g of the fertilizer and dissolve it in about 50 cm^3 of distilled water.
Step 2 Adding $MgSO_4(aq)$	Add about 50 cm^3 of aqueous solution of magnesium sulphate.
Step 3 Adding $NH_3(aq)$	Slowly add aqueous ammonia with stirring until no more precipitate is formed.
Step 4 Filtering off the precipitate	Filter off the precipitate under reduced pressure through a piece of weighed filter paper. Rinse the beaker with ice-cold distilled water and pour all the washings over the precipitate.
Step 5 Washing the precipitate	Wash the precipitate with several 2 cm^3 portions of propan-2-ol.
Step 6 Allowing the precipitate to dry	Allow the precipitate to dry with the water aspirator still operating. Then remove the precipitate with the filter paper from the funnel. Place it on a watch glass for drying overnight.
Step 7 Weighing the precipitate and the filter paper	Weigh the precipitate and the filter paper the day after.

Extension

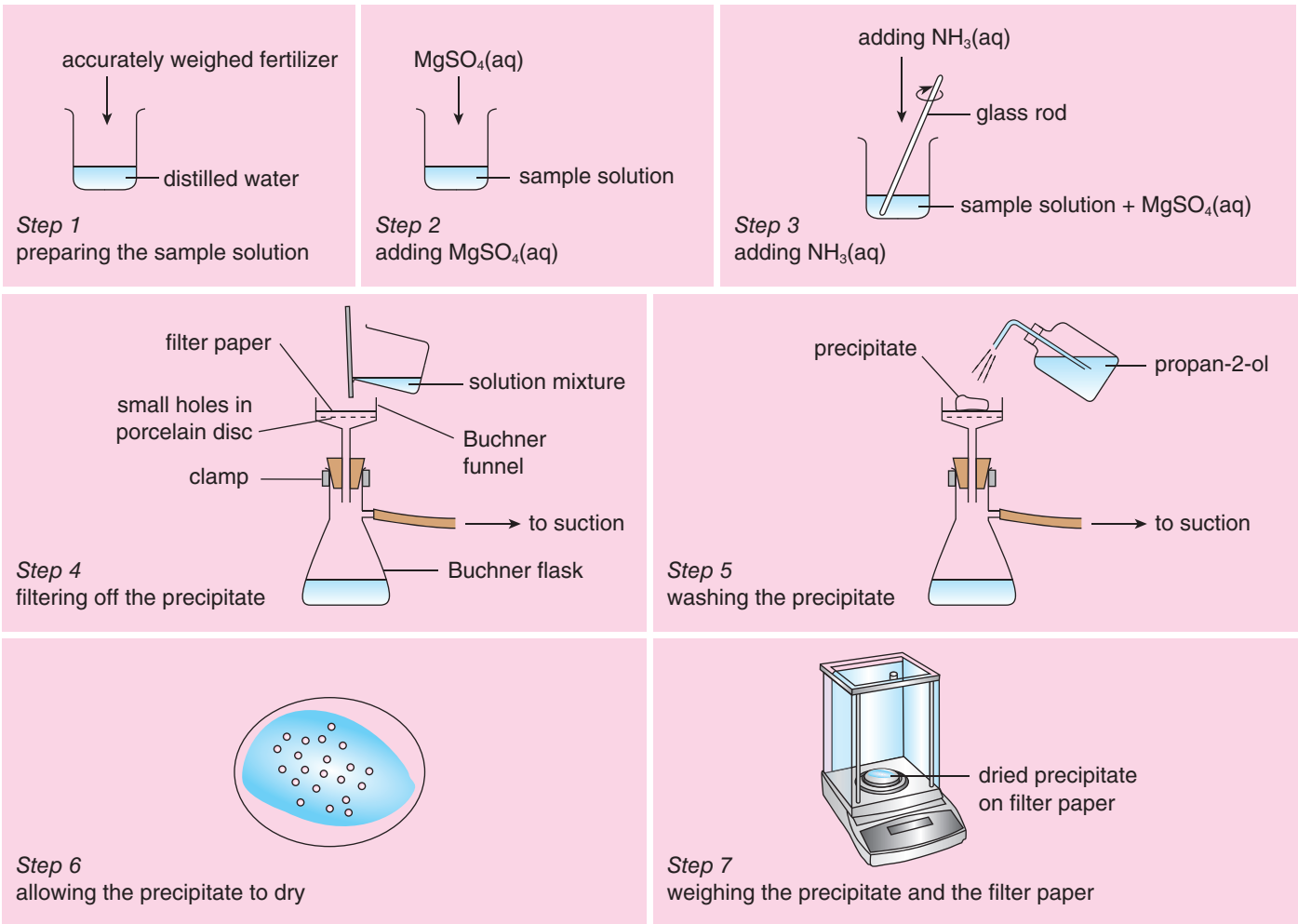


Fig. 55.7 Determining the phosphorus content in a sample of fertilizer

Calculating the percentage by mass of phosphorus in the sample of fertilizer

The following results are obtained in a certain experiment:

Mass of fertilizer used = 3.050 g

Mass of dry filter paper = 1.234 g

Mass of filter paper + precipitate = 3.061 g

Given that:

molar mass of P = 31.0 g mol^{-1},

molar mass of P_2O_5 = 142.0 g mol^{-1}, and

molar mass of $MgNH_4PO_4 \cdot 6H_2O$ = 245.3 g mol^{-1}

Extension

We can calculate the percentage by mass of phosphorus in the sample of fertilizer as follows:

sample of fertilizer $\longrightarrow$ $MgNH_4PO_4 \bullet 6H_2O$
3.050 g (3.061 – 1.234) g = 1.827 g

$$\text{Number of moles of } MgNH_4PO_4 \bullet 6H_2O = \frac{1.827 \text{ g}}{245.3 \text{ g}}$$
$$= 0.007448 \text{ mol}$$

According to the formula of the precipitate, $MgNH_4PO_4 \bullet 6H_2O$, 1 mole of P gives 1 mole of the precipitate.

i.e. number of moles of P in sample = 0.007448 mol

$$\text{Mass of P in sample} = 0.007448 \text{ mol} \times 31.0 \text{ g mol}^{-1}$$
$$= 0.2309 \text{ g}$$

$$\text{\% by mass of P in sample} = \frac{0.2309 \text{ g}}{3.050 \text{ g}} \times 100\%$$
$$= 7.570\%$$

To compare the experimental result with the label on the fertilizer, we need to convert the percentage by mass of phosphorus to percentage by mass of phosphorus pentoxide.

$$\text{Mass of } P_2O_5 \text{ in sample} = \frac{0.007448}{2} \text{ mol} \times 142.0 \text{ g mol}^{-1}$$
$$= 0.5288 \text{ g}$$

$$\text{Percentage by mass of } P_2O_5 \text{ in sample} = \frac{0.5288 \text{ g}}{3.050 \text{ g}} \times 100\%$$
$$= 17.34\%$$

Therefore the percentage by mass of phosphorus in the fertilizer is 7.570% and that of phosphorus pentoxide is 17.34%.

Fig. 55.8 summarizes the steps in calculating the percentage by mass of the species being analyzed in gravimetric analysis involving a precipitation reaction.

Extension

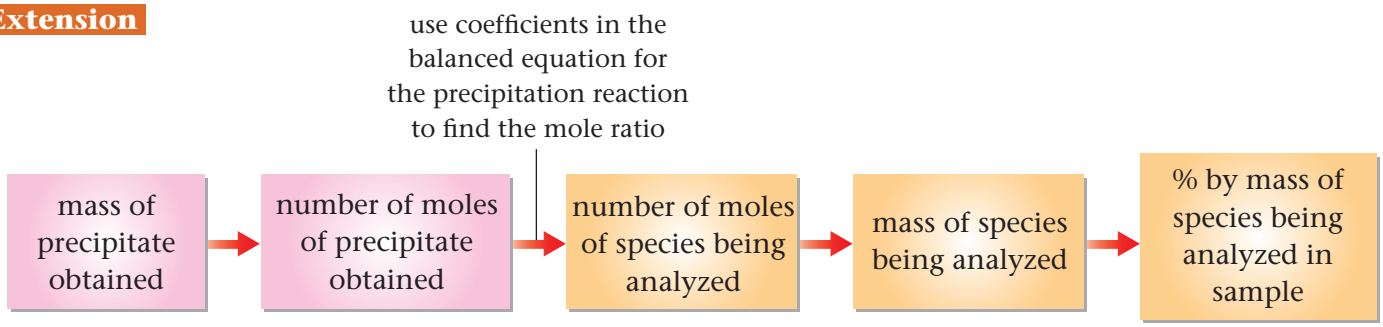


Fig. 55.8 Steps in calculating the percentage by mass of the species being analyzed in gravimetric analysis involving a precipitation reaction

Sources of error

The following sources of error make the experimental result for the percentage by mass of phosphorus in the fertilizer slightly lower than the true result:

- not all phosphorus is precipitated out as we may not have added enough precipitating agent*;
- as no ionic substance is completely insoluble in water, a little magnesium ammonium phosphate will remain dissolved in the solution;
- a little of the precipitate will be lost as we wash the precipitate;
- little splashes, inefficient rinsing out of the beaker, and inefficient filtering can cause some loss of the precipitate.

The following sources of error make the experimental result for the percentage by mass of phosphorus in the fertilizer slightly higher than the true result:

- impurities may be present;
- the precipitate may not be dried completely and so residual moisture can add to its mass.

* To test this, add a little precipitating agent to the mother liquor after filtering off the precipitate. If some phosphorus is still present, a precipitate will form. Then add more precipitating agent to precipitate the remaining phosphorus and re-filter.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Refer to page T58.

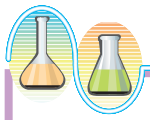
Suggest possible ways to minimize possible sources of error in the experiment.

Extension

In the following activity, you are going to determine the phosphorus content in a sample of fertilizer.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.



Activity 55.1

Determining the phosphorus content in a sample of fertilizer.

55.5 Gravimetric determination of calcium content in a sample solution

We can determine the calcium content in a sample solution by precipitation with oxalate ions ($C_2O_4^{2-}$).

$$Ca^{2+}(aq) + C_2O_4^{2-}(aq) + H_2O(l) \longrightarrow \underset{\text{calcium oxalate}}{CaC_2O_4 \bullet H_2O(s)}$$

Suppose we need to determine the mass of calcium ions in 100.0 cm^3 of a sample solution. During the experiment, add an aqueous solution of ammonium oxalate ($(NH_4)_2C_2O_4$) and solid urea ($CO(NH_2)_2$) to the acidified sample solution. The following equilibrium exists in the reaction mixture:

$$(NH_4)_2C_2O_4(aq) + 2H^+(aq) \rightleftharpoons 2NH_4^+(aq) + H_2C_2O_4(aq)$$

Notice that the precipitating agent is oxalate ion. Very little precipitate will form at this stage as the oxalate ions exist mainly in the form of oxalic acid ($H_2C_2O_4$).

However, upon heating, the urea slowly decomposes to form hydroxide ions, OH^-. The hydroxide ions neutralize the oxalic acid to liberate oxalate ions. Precipitation occurs slowly allowing for slow crystal growth, hence larger and purer crystals develop.

$$CO(NH_2)_2(s) + 3H_2O(l) \longrightarrow 2NH_4^+(aq) + CO_2(g) + 2OH^-(aq)$$

$$2OH^-(aq) + H_2C_2O_4(aq) \longrightarrow C_2O_4^{2-}(aq) + 2H_2O(l)$$

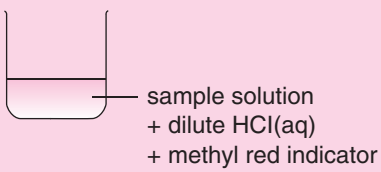
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Extension

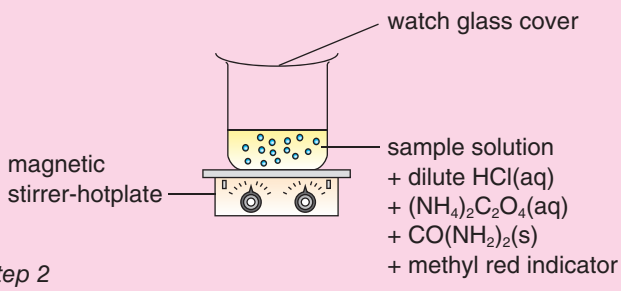
* This indicator is red below pH 4.8 and yellow above pH 6.0.

The experimental details are as follows:

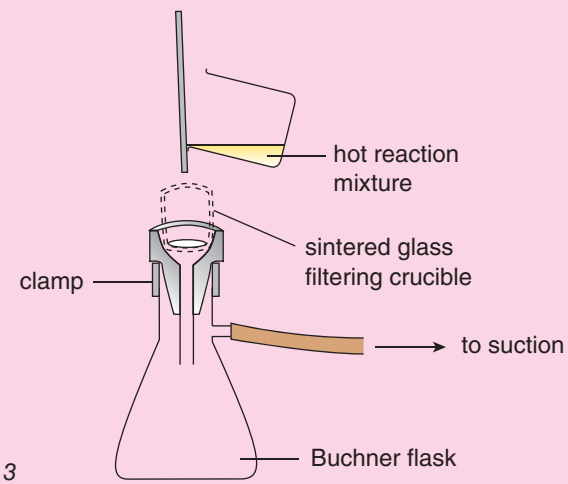
Step 1 Preparing the sample solution	• Pipette 25.0 cm^3 of the sample solution to a beaker. • Dilute the sample solution with about 75 cm^3 of 0.1 mol dm^{-3} hydrochloric acid. • Add 5 drops of methyl red indicator*.
Step 2 Precipitating the calcium oxalate	• Add 25 cm^3 of aqueous solution of ammonium oxalate. • Add 15 g of solid urea. • Boil gently for about 30 minutes until the indicator turns yellow.
Step 3 Filtering off the precipitate	• Filter the hot reaction mixture under reduced pressure through a weighed sintered glass filtering crucible.
Step 4 Washing the precipitate	• Rinse the beaker with ice-cold distilled water and pour all the washings over the precipitate. Wash the precipitate with ice-cold distilled water.
Step 5 Drying the precipitate	• Dry the precipitate in an oven at 105 °C for 1 hour. • Store the precipitate in a desiccator while cooling.
Step 6 Weighing the precipitate and the filtering crucible	• The product is hygroscopic. Therefore weigh the precipitate and the filtering crucible rapidly.



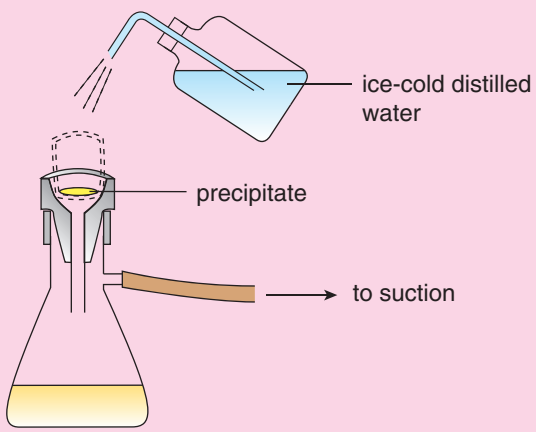
Step 1
preparing the sample solution



Step 2
precipitating the calcium oxalate

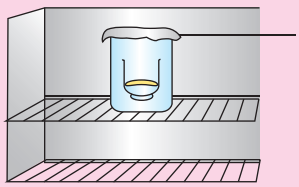


Step 3
filtering off the precipitate

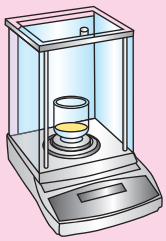


Step 4
washing the precipitate

Extension



Step 5
drying the precipitate*



Step 6
weighing the precipitate and the filtering crucible

Fig. 55.9 Steps in determining the calcium content in a sample solution by precipitation with oxalate ion ($C_2O_4^{2-}$)

* Place the filtering crucible in a beaker before drying in the oven.

Calculating the percentage by mass of calcium ions in the sample

The following results are obtained in the experiment:

Volume of sample solution used = 25.0 cm^3

Mass of filtering crucible = 26.600 g

Mass of filtering crucible plus precipitate = 27.054 g

Given that:

molar mass of Ca = 40.1 g mol^{-1}

molar mass of $CaC_2O_4 \bullet H_2O$ = 146.1 g mol^{-1}

We can calculate the mass of calcium ions per 100.0 cm^3 of the sample solution as follows:

$Ca^{2+}(aq)$ +	$C_2O_4^{2-}(aq)$ + $H_2O(l)$ $\rightleftharpoons$	$CaC_2O_4 \bullet H_2O(s)$
in 25.0 cm^3 of sample solution		(27.054 – 26.600) g = 0.454 g

$$\text{Number of moles of } CaC_2O_4 \bullet H_2O = \frac{0.454 \text{ g}}{146.1 \text{ g mol}^{-1}} = 0.00311 \text{ mol}$$

Number of moles of Ca^{2+} ions in 25.0 cm^3 of sample solution = 0.00311 mol

Mass of Ca^{2+} ions in 25.0 cm^3 of sample solution
= 0.00311 mol x 40.1 g mol^{-1}
= 0.125 g

Extension

Mass of Ca^{2+} ions in 100.0 cm^3 of sample solution

$$= \frac{0.125\ g}{25.0\ cm^3} \times 100.0\ cm^3$$

$$= 0.500\ g$$

Therefore the sample solution contains 0.500 g of calcium ions per 100.0 cm^3.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Refer to page T58.

Several actions which could occur in different steps during this analytical procedure are listed below. For each action, indicate the likely effect on calculated mass of calcium ions per 100.0 cm^3 of the sample solution by placing a tick (✔) in the appropriate box. Explain your answer in each case.

Action		Calculated result would be lower than the true value	No effect on calculated result	Calculated result would be higher than the true value
Step 1	The beaker for holding the 25.0 cm^3 of sample solution had been previously washed with distilled water but not dried.		✔	
Step 3	The mass of the empty filtering crucible was recorded incorrectly. The recorded mass was 0.100 g higher than the actual mass.	✔		
Step 4	The beaker containing the reaction mixture was not rinsed after pouring.	✔		
Step 4	The precipitate was not washed.			✔
Step 5	The dried precipitate was not kept in a desiccator while cooling.			✔

In the following activity, you are going to determine the calcium content in a sample solution.

Extension

Video

Filtration, drying and weighing the precipitate obtained in gravimetric analysis

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 55.2

Determining the calcium content in a sample solution.

Checkpoint

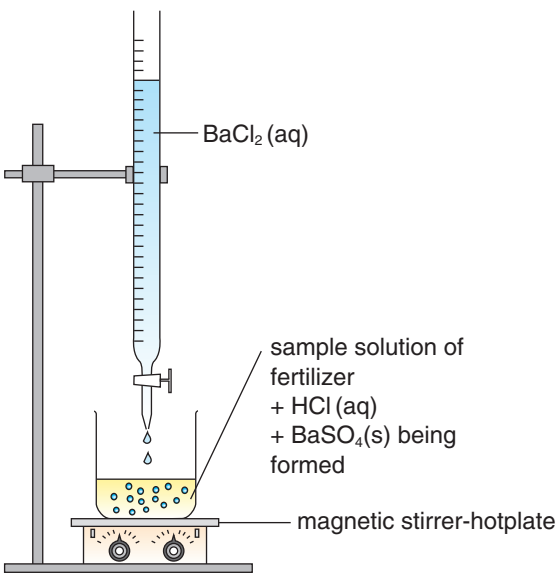
Refer to page T58.

1 A certain brand of foot powder contains zinc. The zinc in a 1.080 g sample of the powder is precipitated as the salt $ZnNH_4PO_4$. This salt is then strongly heated, producing 0.368 g of $Zn_2P_2O_7$. Calculate the percentage by mass of zinc in the foot powder. 9.35%

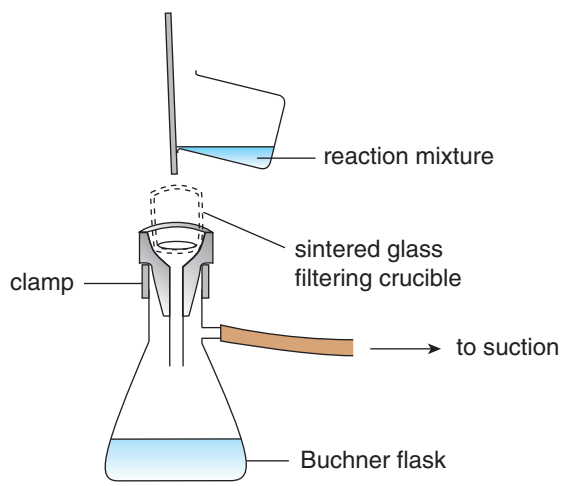
(Relative atomic masses: O = 16.0, P = 31.0, Zn = 65.4)

2 A student determined the sulphate content of a sample of fertilizer using the following procedure:

Step 1	Weigh out a sample of the fertilizer and transfer it to a beaker. Add a little concentrated hydrochloric acid and gently heat it to boiling.
Step 2	Set up a burette containing dilute aqueous solution of barium chloride above the beaker. Add the solution dropwise until no more white precipitate forms.
Step 3	Use a weighed sintered glass filtering crucible to filter off the barium sulphate precipitate under reduced pressure.
Step 4	Wash the precipitate with hot water and ethoxyethane.
Step 5	Dry and weigh the precipitate.



Apparatus for the precipitation of barium sulphate



Filtration using a sintered glass filtering crucible

Extension

The results of the experiment are as follows:

Mass of sample of fertilizer = 0.220 g

Mass of sintered glass filtering crucible = 18.456 g

Mass of filtering crucible + $BaSO_4$ precipitate = 18.791 g

a) Write an ionic equation for the precipitation reaction occurred.

b) Suggest how you could show that complete precipitation has occurred in *Step 2*.

c) Why is the precipitate washed before drying?

d) Calculate the percentage by mass of sulphate in the fertilizer. 62.7%

e) Assuming that all the sulphate is derived from ammonium sulphate, determine the percentage by mass of ammonium sulphate in the fertilizer. 86.4%

f) What are the sources of error in the experiment?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, S = 32.1, Ba = 137.3)

55.6 Precipitation titrations in volumetric analysis

Titrations are widely used in analytical chemistry. Precipitation titrations are based upon reactions that yield insoluble ionic compounds. The most important precipitating agent is silver nitrate.

For example, we can determine the concentration of chloride ions in a sample solution by titrating it with a standard aqueous solution of silver nitrate. As we add the aqueous solution of silver nitrate slowly to the sample solution, a precipitate of silver chloride forms.

$$Ag^+(aq) + Cl^-(aq) \rightleftharpoons AgCl(s)$$

We can use a chromate indicator in the titration. When all the chloride ions are precipitated, the first excess silver nitrate solution gives a reddish brown silver chromate precipitate with the chromate indicator. This signals the end point of the titration.

$$2Ag^+(aq) + CrO_4^{2-}(aq) \rightleftharpoons \underset{\text{reddish brown}}{Ag_2CrO_4(s)}$$

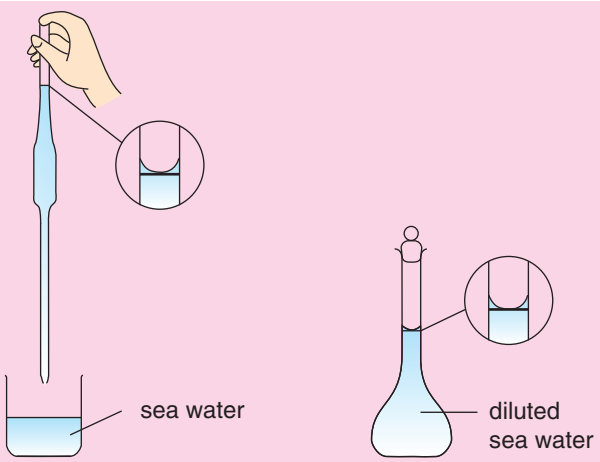
We can use this method to determine the concentration of chloride ions of water samples from many sources such as sea water, stream water and river water.

The titration must be performed at a pH of about 8. If the solution is too acidic (pH < 6), then part of the indicator will be present as hydrogen chromate ions ($HCrO_4^-$) and more silver ions will be required to form the silver chromate precipitate. Above pH 8, silver oxide may be precipitated. We can maintain the proper pH by adding solid sodium hydrogencarbonate to the solution.

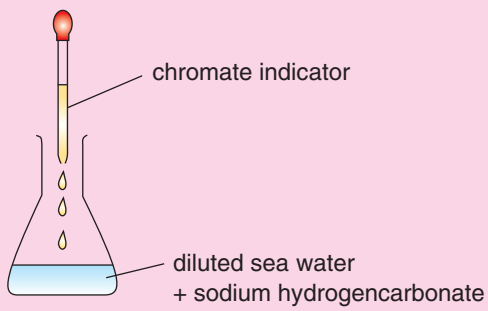
This titration is sensitive to the presence of both chloride ions and bromide ions in the sample solution. Therefore it will not be too accurate when there is a significant concentration of bromide ions present as well as the chloride ions. However, in most cases, such as in sea water, the concentration of bromine ions will be negligible.

Determining the concentration of chloride ions in sea water by precipitation titration

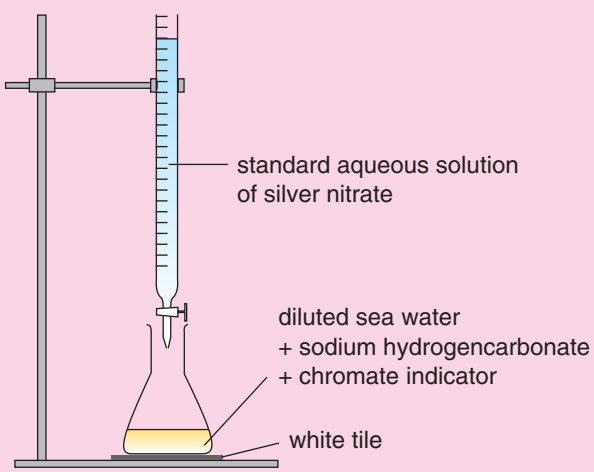
To determine the concentration of chloride ions in sea water, follow the steps listed below:



Step 1
Pipette 25.0 cm^3 of the sea water to a 250.0 cm^3 volumetric flask. Dilute to make 250.0 cm^3 dilute sea water with distilled water.



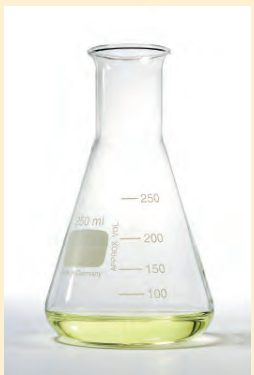
Step 2
Place 25.0 cm^3 of the diluted sea water in a conical flask using the pipette. Add a little sodium hydrogencarbonate and the chromate indicator.



Step 3
Run the standard aqueous solution of silver nitrate into the diluted sea water until a reddish brown colour appears.

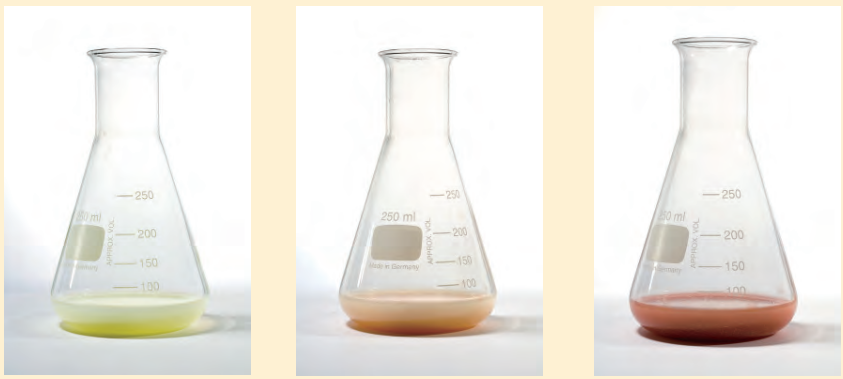
Fig. 55.10 Steps in determining the concentration of chloride ions in sea water

a)



A clear solution of yellow colour given by the chromate indicator before the addition of any aqueous solution of silver nitrate

b)



Left: before the titration end point, AgCl(s) forms and gives a cloudy mixture

Middle: at the end point, all Cl^- ions have precipitated and the first excess $AgNO_3(aq)$ gives a reddish brown precipitate with the chromate indicator

Right: beyond the end point, further addition of $AgNO_3(aq)$ forms more precipitate and gives a deeper reddish brown colour

Fig. 55.11 Changes in the reaction mixture during the precipitation titration of chloride ions in sea water

Calculating the concentration of chloride ions in a sample of sea water

In a certain experiment, a 25.0 cm^3 sample of sea water was diluted to make 250.0 cm^3 of solution. 25.0 cm^3 of the diluted sea water required 17.6 cm^3 of 0.0800 mol dm^{-3} aqueous solution of silver nitrate for complete reaction.

We can calculate the concentration of chloride ions in the sample as follows:

$$\underset{\substack{0.0800\ \text{mol dm}^{-3}\\ 17.6\ \text{cm}^3}}{Ag^+(aq)} + \underset{\substack{25.0\ \text{cm}^3\\ \downarrow\\ 250.0\ \text{cm}^3\\ \downarrow\\ \text{(used) } 25.0\ \text{cm}^3}}{Cl^-(aq)} \longrightarrow AgCl(s)$$

Number of moles of Ag^+ ions in 17.6 cm^3 solution

$$= 0.0800\ \text{mol dm}^{-3} \times \frac{17.6}{1\ 000}\ \text{dm}^3$$

$$= 0.00141\ \text{mol}$$

According to the equation, 1 mole of Ag^+ ions reacts with 1 mole of Cl^- ions.

i.e. number of moles of Cl^- ions in 25.0 cm^3 diluted sea water
= 0.00141 mol

Number of moles of Cl^- ions in 250.0 cm^3 diluted sea water
= 10 x 0.00141 mol
= 0.0141 mol
= number of moles of Cl^- ions in 25.0 cm^3 sea water

Concentration of Cl^- ions in sea water

$$= \frac{0.0141 \text{ mol}}{\frac{25.0}{1\ 000}\ dm^3}$$

$= 0.564 \text{ mol } dm^{-3}$
$= 0.564 \text{ mol } dm^{-3} \times 35.5 \text{ g } mol^{-1}$*
$= 20.0 \text{ g } dm^{-3}$

* Molar mass of Cl
= 35.5 g mol^{-1}

Therefore the concentration of chloride ions in the sample of sea water is 20.0 g dm^{-3}.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

List possible sources of error for the above experiment.

Discussion
Some sources of error students may suggest:
- Bromide ions might be present in sea water.
- The burette, pipette and volumetric flask might hold or deliver volumes slightly different from the volumes indicated by the manufacturer.
- The level of eyes might not be on the same level as the graduation marks of the burette, pipette and volumetric flask when reading the graduation marks and thus leads to reading errors.
- It was difficult to tell exactly when the indicator changed colour.
- The aqueous solution of silver nitrate might be added too fast, not allowing enough time for the reaction between silver ions and chloride ions to take place. This would cause the addition of more aqueous solution of silver nitrate than required.
- Deciding when the titration end point is reached requires personal judgement and thus may lead to errors.
- Little splashes can cause errors.

In the following activity, you are going to determine the concentration of chloride ions in a sea water sample as discussed above.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 55.3

Determining the concentration of chloride ions in a sea water sample by titration with a standard aqueous solution of silver nitrate.

Extension

Gravimetric determination of the concentration of chloride ions in a sample solution by precipitation

As discussed earlier, we can also determine the concentration of chloride ions in a sample solution by gravimetric analysis. We can precipitate silver chloride by adding an aqueous solution of silver nitrate to the sample solution.

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

The precipitate is collected by careful filtration and weighed. As this method requires very careful weighing, it is best to use it on solutions containing a fairly significant concentration of chloride ions, such as sea water.

55.7 Redox titrations in volumetric analysis

We have seen in Topic 4 Acids and Bases that we can determine the concentration of an acid or alkaline solution by titration. We can carry out a similar procedure to determine the concentration of an oxidizing agent or a reducing agent — a process called **redox titration**. All that is necessary is that the species being analyzed undergoes an oxidation or a reduction process completely and that there be some means, such as a colour change, to tell when the reaction is complete.

In this unit, we are going to discuss redox titrations involving the following two reagents:

- potassium permanganate; and
- iodine.

55.8 Redox titrations involving an aqueous solution of potassium permanganate

Potassium permanganate ($KMnO_4$) is a strong oxidizing agent with an intense purple colour. In an acidic solution, it is reduced to colourless manganese(II) ions (Mn^{2+}).

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$$

redox titration 氧化還原滴定法

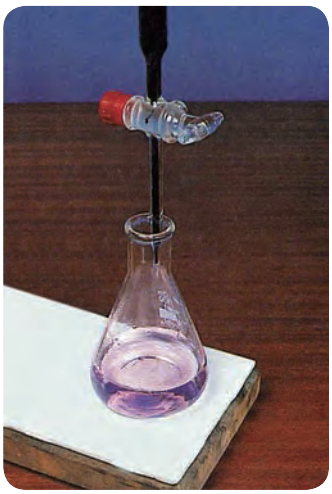


Fig. 55.12 In a redox titration involving an aqueous solution of potassium permanganate, one drop of excess permanganate solution colours the contents of the conical flask pale pink

In neutral and alkaline solutions, the product is the brown solid, manganese(IV) oxide (MnO_2).

ASL 2001 Q3(c)
AL 2002 Paper 1 Q7(a)(i)

$$MnO_4^-(aq) + 2H_2O(l) + 3e^- \longrightarrow MnO_2(s) + 4OH^-(aq)$$

For a titration in an acidic solution, an aqueous solution of potassium permanganate serves as its own indicator because the product, manganese(II) ion, is colourless. One drop of excess permanganate solution colours the contents of the conical flask pale pink (Fig. 55.12). We can take the appearance of such a colour as the end point.

Due to its strong oxidizing power, an aqueous solution of potassium permanganate oxidizes the chloride ions to chlorine. This limits the applications of the reagent. For example, we cannot use the reagent to analyze the iron content in a sample of iron(II) chloride.

Standardization of aqueous solution of potassium permanganate

ASL 2011 Paper 1 Q1(c)
AL 2011 Paper 1 Q1(a)(iii)

Traces of manganese(IV) oxide are present in solid potassium permanganate. Therefore it is not feasible to prepare a standard aqueous solution of potassium permanganate by dissolving a known mass of the solid in water.

An aqueous solution of potassium permanganate is unstable as it tends to decompose slowly at room temperature:

$$4MnO_4^-(aq) + 2H_2O(l) \longrightarrow 4MnO_2(s) + 3O_2(g) + 4OH^-(aq)$$

Therefore it is necessary to **standardize** a permanganate solution to obtain its accurate concentration prior to its use for titrimetric analysis.

We can standardize an aqueous solution of potassium permanganate by titrating it against an aqueous solution of sodium oxalate ($Na_2C_2O_4$)*. In an acidic solution, the oxalate ions are converted to oxalic acid ($H_2C_2O_4$)*. Potassium permanganate reacts with the oxalic acid according to the following equation:

ASL 2011 Paper 1 Q1(a)
AL 2011 Paper 1 Q1(a)(i)

$$5H_2C_2O_4(aq) + 2MnO_4^-(aq) + 6H^+(aq) \longrightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

Follow the steps below in the standardization process:

- Dissolve a known mass of sodium oxalate in dilute sulphuric acid.
- Heat the oxalate solution to 80 – 90 °C and titrate with the aqueous solution of potassium permanganate until a pale pink colour persists, showing that all the oxalic acid has reacted.

* Anhydrous sodium oxalate is available in high purity and is stable in air.

* The systematic name of oxalic acid is ethanedioic acid.

standardize 標定

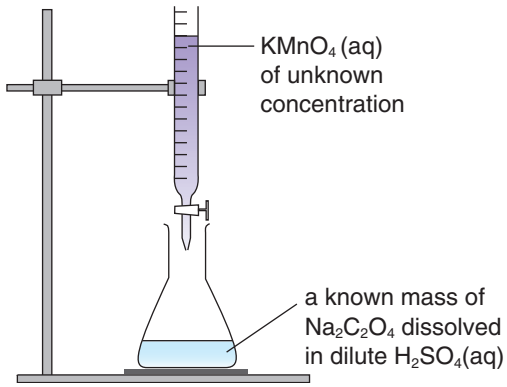


Fig. 55.13 Standardization of an aqueous solution of potassium permanganate

Fig. 55.14 summarizes the steps we can use in calculating the molarity of the aqueous solution of potassium permanganate. We can apply these steps in calculations of similar problems.

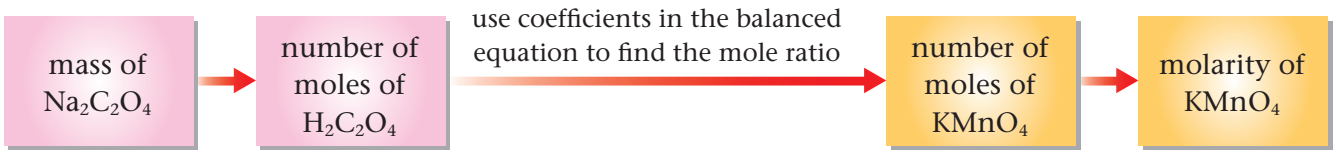


Fig. 55.14 Steps for calculation in $KMnO_4$ standardization

55.9 Determining the iron content in commercial iron tablets



Fig. 55.15 Iron tablets

Iron is essential to the human body. It is a constituent of haemoglobin. Iron is also present in a number of enzymes involved in the redox processes in the body. People who are iron deficient are often prescribed iron tablets by their physicians.

Iron tablets bought at the chemist usually contain iron(II) sulphate. We can determine the amount of iron(II) sulphate in an iron tablet by titrating it against a standard aqueous solution of potassium permanganate.

The iron(II) ions in iron(II) sulphate react with the permanganate ions in an acidic solution according to the following equation:

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$$

ASL 2011 Paper 1 Q1(a)
AL 2011 Paper 1 Q1(a)(i)



55.1 Determining the iron content in a brand of iron tablets

ASL 2001 Q3(b)
AL 2002 Paper 1 Q7(a)(ii)

The following experiment is carried out to determine the mass of iron(II) sulphate in a brand of iron tablets.

Step 1	Grind 2 iron tablets.
Step 2	Dissolve the powder in 50 cm^3 of dilute sulphuric acid. Make up the solution to 250.0 cm^3.
Step 3	Transfer 25.0 cm^3 of the solution to a conical flask.
Step 4	Titrate the solution against a 0.00500 mol dm^{-3} aqueous solution of potassium permanganate. 9.20 cm^3 of the permanganate solution are required to reach the end point.

a) Calculate the mass of iron(II) sulphate in one iron tablet.

b) Could hydrochloric acid be used instead of sulphuric acid in *Step 2*? Explain your answer.

Solution

ASL 2011 Paper 1 Q1(b)
AL 2011 Paper 1 Q1(a)(ii)

a)

$$\underset{\substack{0.00500\ \text{mol}\ dm^{-3}\\ 9.20\ cm^3}}{MnO_4^-(aq)} + 8H^+(aq) + \underset{\substack{2\ \text{iron tablets}\\ \downarrow\\ 250.0\ cm^3\\ \downarrow\\ (\text{used})\ 25.0\ cm^3}}{5Fe^{2+}(aq)} \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$$

Number of moles of MnO_4^- ions in 9.20 cm^3 solution $= 0.00500\ \text{mol}\ dm^{-3} \times \frac{9.20}{1\ 000}\ dm^3$

$= 4.60 \times 10^{-5}$ mol

According to the equation, 1 mole of MnO_4^- ions reacts with 5 moles of Fe^{2+} ions.

i.e. number of moles of Fe^{2+} ions in 25.0 cm^3 solution $= 5 \times 4.60 \times 10^{-5}$ mol

$= 2.30 \times 10^{-4}$ mol

Number of moles of Fe^{2+} ions in 250.0 cm^3 solution

$= 10 \times 2.30 \times 10^{-4}$ mol

$= 0.00230$ mol

= number of moles of Fe^{2+} ions in 2 iron tablets

Number of moles of Fe^{2+} ions in 1 iron tablet $= \frac{0.00230}{2}$ mol

$= 0.00115$ mol

Mass of $FeSO_4$ in 1 iron tablet $= 0.00115$ mol $\times$ 151.9 g mol^{-1}*

$= 0.175$ g

∴ 1 iron tablet contains 0.175 g of iron(II) sulphate.

* Molar mass of $FeSO_4$ = 151.9 g mol^{-1}

b) Not appropriate as the permanganate ions may oxidize the chloride ions in the solution.

In the following activity, you are going to determine the percentage by mass of iron(II) sulphate in a brand of iron tablets.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Video

Determining the percentage by mass of iron(II) sulphate in a brand of iron tablets

Activity 55.4

Determining the percentage by mass of iron(II) sulphate in a brand of iron tablets.

55.10 Analyzing the quality of water by determining its permanganate index

The *permanganate index of water* indicates the amount of permanganate consumed when the oxidizable matter in a water sample is oxidized by the permanganate. The index is a measure of the contamination by organic and oxidizable inorganic matter in a water sample. The amount of permanganate consumed in oxidizing the oxidizable matter in a water sample is usually converted to the equivalent amount of oxygen. The index is expressed in milligrams of oxygen per litre (dm^3) of water.

The permanganate index is intended for judging the quality of potable water and raw waters, such as well and surface waters. We may need to do an appropriate dilution when analyzing heavily contaminated waters.

permanganate index of water 水的高錳酸鹽指數

What is the equivalence between permanganate and oxygen?

When the permanganate oxidizes the oxidizable matter in water under acidic conditions, it undergoes the following reduction process:

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$$

If oxygen is used for the oxidation, it will undergo the following process:

$$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$$

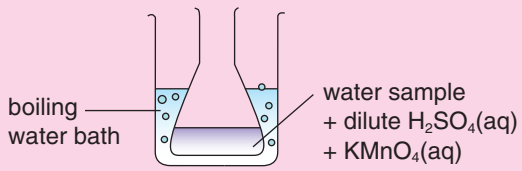
Since each MnO_4^- ion consumes 5 electrons and each O_2 molecule consumes 4 electrons, then one mmole (millimole) of MnO_4^- ions will oxidize the same amount of matter as 1.2 mmoles of O_2.

Thus, from the amount of permanganate consumed in oxidizing the matter in a certain volume of the water sample, we can calculate the permanganate index of the water sample.

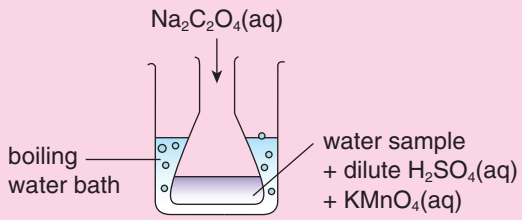
We can determine the permanganate index of a water sample using the steps listed below. The following two reagents are used:

- aqueous solution of potassium permanganate; and
- aqueous solution of sodium oxalate.

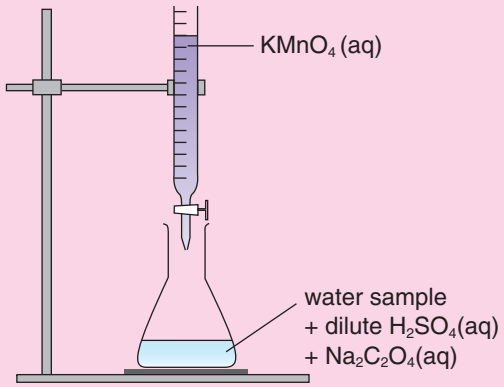
Step 1	Allow 25.0 cm^3 of the water sample to react with 5.00 cm^3 of aqueous solution of potassium permanganate under acidic conditions. Heat the mixture in a boiling water bath. A portion of the permanganate will be consumed for oxidizing the matter in the water sample.
Step 2	Add 5.00 cm^3 of aqueous solution of sodium oxalate to react with the permanganate that remains in the mixture.
Step 3	Determine the oxalate remaining in the mixture by titration against aqueous solution of potassium permanganate.
Step 4	Determine the quantity of permanganate consumed for oxidizing the matter in the water sample.
Step 5	Calculate the permanganate index of the water sample in mg of O_2 per litre of water.



Step 1
allowing the water sample to react with aqueous solution of potassium permangante under acidic conditions



Step 2
adding aqueous solution of sodium oxalate



Step 3
determining the oxalate remaining in the mixture by titration against aqueous solution of potassium permanganate

Fig. 55.16 Steps in determining the permanganate index of a water sample

Calculating the permanganate index of a water sample

The following results are obtained in a certain experiment:

Concentration of $KMnO_4$(aq) = 0.00200 mol dm^{-3}

Concentration of $Na_2C_2O_4$(aq) = 0.00500 mol dm^{-3}

Volume of $KMnO_4$(aq) required to react with the oxalate remaining in the mixture = 3.50 cm^3

Under acidic conditions, the oxalate ions are converted to oxalic acid. Potassium permanganate reacts with the oxalic acid according to the following equation:

$$\underset{\substack{0.00500\ \text{mol}\ dm^{-3}\\ 5.00\ cm^3}}{5H_2C_2O_4(aq)} + \underset{0.00200\ \text{mol}\ dm^{-3}}{2MnO_4^-(aq)} + 6H^+(aq) \longrightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

Number of moles of $C_2O_4^{2-}$ ions added in *Step 2*

$$= 0.00500\ \text{mol}\ dm^{-3} \times \frac{5.00}{1\ 000}\ dm^3$$

$$= 2.50 \times 10^{-5}\ \text{mol}$$

According to the equation, 5 moles of $C_2O_4^{2-}$ ions react with 2 moles of MnO_4^- ions.

i.e. number of moles of $KMnO_4$ required to react with the $C_2O_4^{2-}$ ions $= \frac{2}{5} \times 2.50 \times 10^{-5}$ mol

$= 1.00 \times 10^{-5}$ mol

Volume of $KMnO_4$ required to react with $C_2O_4^{2-}$ ions

$= \frac{1.00 \times 10^{-5} \text{ mol}}{0.00200 \text{ mol dm}^{-3}}$

$= 5.00 \times 10^{-3}$ dm^3

$= 5.00$ cm^3

Notice that 5.00 cm^3 of $KMnO_4$(aq) and 5.00 cm^3 of $Na_2C_2O_4$(aq) are added to the water sample in *Steps 1* and *2*. If none of the permanganate ion is consumed by the oxidizable matter in the water sample, all the oxalate ions will react with the permanganate ions and no oxalate ion will remain in the mixture. As 3.50 cm^3 of aqueous solution of permanganate are required to react with the oxalate ions remaining in the mixture, we can deduce that this amount of permanganate ions is consumed in oxidizing the oxidizable matter in the water sample.

Number of moles of MnO_4^- ions consumed in oxidizing matter in water sample $= 0.00200 \text{ mol dm}^{-3} \times \frac{3.50}{1\ 000}$ dm^3

$= 7.00 \times 10^{-6}$ mol

$= 7.00 \times 10^{-3}$ mmol

Number of millimoles of O_2 required to oxidize the same amount of matter $= 1.2 \times 7.00 \times 10^{-3}$ mmol

$= 8.40 \times 10^{-3}$ mmol

Mass of O_2 required to oxidize the same amount of matter

$= 8.40 \times 10^{-3}$ mmol $\times$ 32.0 mg mmol^{-1}

$= 0.269$ mg

Permanganate index of water sample

$= \frac{0.269 \text{ mg}}{\frac{25.0}{1\ 000} \text{ L}}$

$= 10.8$ mg of O_2 per litre

In the following activity, you are going to determine the permanganate index of a water sample.

Video
Determining the permanganate index of a water sample

Activity 55.5

Determining the permanganate index of a water sample.

Checkpoint

Refer to page T60.

1 An experiment is carried out to determine the percentage by mass of iron in an iron wire. The experiment consists of two stages.

Stage 1	Add a 0.190 g sample of the iron wire to excess dilute sulphuric acid. Iron(II) sulphate solution results.
Stage 2	Titrate the iron(II) sulphate solution against 0.0215 mol dm^{-3} aqueous solution of potassium permanganate. 28.8 cm^3 of the permanganate solution are required to reach the end point.

a) Write an ionic equation for the reaction taking place during the titration.

b) Suggest how the end point of the titration can be detected.

c) Calculate the percentage by mass of iron in the sample of the iron wire. 91.1%

d) State ONE assumption of the experiment.

(Relative atomic mass: Fe = 55.8)

ASL 2006 Q9(c)

2 10.0 cm^3 of an aqueous solution of hydrogen peroxide are diluted to 500.0 cm^3 with distilled water. 25.0 cm^3 of this diluted solution are titrated against a 0.0100 mol dm^{-3} aqueous solution of potassium permanganate under acidic conditions. 35.0 cm^3 of the permanganate solution are required to reach the end point.

The reaction between hydrogen peroxide and potassium permanganate can be represented by the following equation:

$$5H_2O_2(aq) + 2MnO_4^-(aq) + 6H^+(aq) \longrightarrow 5O_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

a) Calculate the concentration of the aqueous solution of hydrogen peroxide in g dm^{-3}. 59.5 g dm^{-3}

b) Explain why this method of titration is NOT suitable for determining the concentration of hydrogen peroxide in a rainwater sample.

(Relative atomic masses: H = 1.0, O = 16.0)

55.11 Determining the vitamin C content in a sample using a redox titration involving iodine

Vitamin C is also known as ascorbic acid. It is an antioxidant vitamin needed for the formation of collagen to hold the cells together and for healthy teeth, gums and blood vessels.

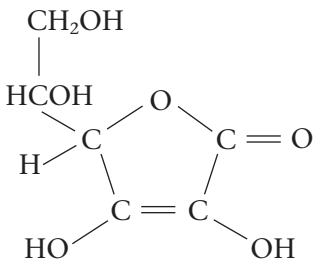


Fig. 55.17 Structure of vitamin C



Fig. 55.18 Some fruits containing vitamin C

We can determine the concentration of a reducing agent in a sample solution by measuring the amount of iodine required to undergo a redox reaction with the reducing agent in a known volume of the sample.

$$\text{reducing agent} + I_2 \longrightarrow I^- + \text{other products}$$

Vitamin C can reduce iodine to iodide ions according to the following equation:

$$\underset{\text{vitamin C}}{C_6H_8O_6(aq)} + I_2(aq) \longrightarrow \underset{\substack{\text{oxidized form}\\ \text{of vitamin C}}}{C_6H_6O_6(aq)} + 2H^+(aq) + 2I^-(aq)$$

Thus, we can determine the vitamin C content in a white grape juice concentrate by measuring the amount of iodine required to undergo a redox reaction with the vitamin C in a known volume of the juice concentrate.

N3 Refer to page T61.

As aqueous iodine solution is unstable, we can generate iodine *in situ* by the reaction of iodate ions (IO_3^-)* with iodide ions in an acidic solution: N3

$$IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \longrightarrow 3I_2(aq) + 3H_2O(l)$$

* IO_3^- ion is also called iodate(V) ion.

AL 2001 Paper 2 Q1(c)(i)
AL 2009 Paper 1 Q7(a)

ASL 2003 Q12(b)(I)(i),(iii)

in situ 原位產生

The experiment consists of two steps:

Step 1	Add aqueous solution of potassium iodide solution, dilute hydrochloric acid and starch solution to 25.0 cm^3 of the white grape juice concentrate.
Step 2	Run the standard aqueous solution of potassium iodate from a burette into the mixture from ***Step 1*** until the first appearance of a persistent dark blue colour.

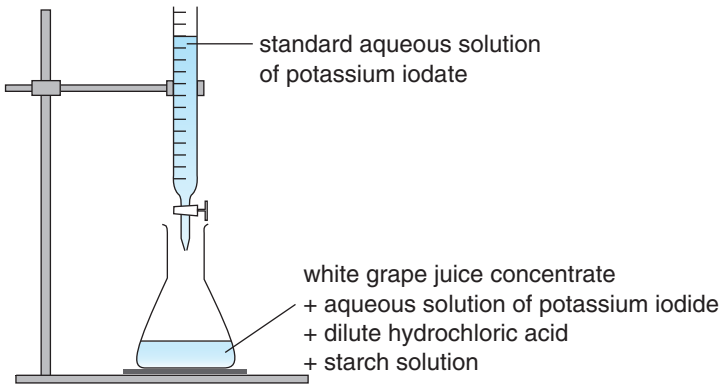


Fig. 55.19 Titration set-up for determining the vitamin C content in a white grape juice concentrate

As the potassium iodate is added to the mixture from *Step 1*, it reacts with the iodide ions to form iodine. The iodine formed immediately oxidizes the vitamin C in the mixture.

Once all the vitamin C is oxidized, there will be an excess of iodine that will react with the starch to form a dark blue complex. This change indicates the end point.

ASL 2003 Q12(b)(I)(ii)
ASL 2006 Q8(I)(c)(ii)

Calculating the vitamin C content in the white grape juice concentrate

Fig. 55.20 summarizes the steps we can follow to calculate the vitamin C content in the white grape juice concentrate.

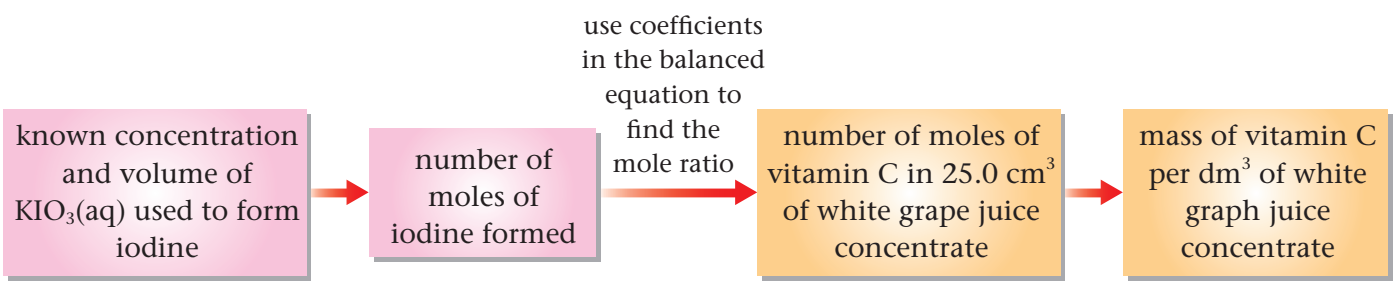


Fig. 55.20 Steps for calculating the vitamin C content in a white grape juice concentrate

The following results are obtained in a certain experiment:

Concentration of $KIO_3(aq)$ = 0.00500 mol dm^{-3}

Volume of $KIO_3(aq)$ required to reach the end point = 22.0 cm^3

We can calculate the vitamin C content of the white grape juice concentrate as follows:

$$\underset{\substack{0.00500 \text{ mol } dm^{-3} \\ 22.0 \text{ cm}^3}}{IO_3^-(aq)} + 5I^-(aq) + 6H^+(aq) \longrightarrow \underset{? \text{ mol}}{3I_2(aq)} + 3H_2O(l)$$

Number of moles of KIO_3 in 22.0 cm^3 solution

$$= 0.00500 \text{ mol } dm^{-3} \times \frac{22.0}{1\ 000} dm^3$$

$$= 1.10 \times 10^{-4} \text{ mol}$$

According to the equation, 1 mole of IO_3^- ions reacts with I^- ions and H^+ ions to give 3 moles of I_2.

i.e. number of moles of I_2 formed = $3 \times 1.10 \times 10^{-4}$ mol

$= 3.30 \times 10^{-4}$ mol

ASL 2003 Q12(b)(I)(i),(iii)

Vitamin C reacts with iodine according to the following equation:

$$\underset{\substack{? \text{ g in } 1.00 \text{ dm}^3 \\ 25.0 \text{ cm}^3}}{C_6H_8O_6(aq)} + \underset{3.30 \times 10^{-4} \text{ mol}}{I_2(aq)} \longrightarrow C_6H_6O_6(aq) + 2H^+(aq) + 2I^-(aq)$$

According to the equation, 1 mole of vitamin C reacts with 1 mole of I_2.

i.e. number of moles of vitamin C in 25.0 cm^3 grape juice concentrate = 3.30×10^{-4} mol

Number of moles of vitamin C per dm^3 of grape juice concentrate

$$= \frac{3.30 \times 10^{-4} \text{ mol}}{\frac{25.0}{1\ 000} dm^3}$$

$$= 0.0132 \text{ mol } dm^{-3}$$

Mass of vitamin C per dm^3 of grape juice concentrate

$= 0.0132 \text{ mol } dm^{-3} \times 176.0 \text{ g } mol^{-1}$*

$= 2.32 \text{ g } dm^{-3}$

Therefore the vitamin C content in the white grape juice concentrate is 2.32 g dm^{-3}.

* Molar mass of vitamin C is 176.0 g mol^{-1}.

55.12 Determining the concentration of sodium hypochlorite in a household bleach using an iodine / thiosulphate titration

To determine the concentration of an oxidizing agent in a sample solution, we can allow the oxidizing agent in a known volume of the sample solution to react with excess iodide ions to liberate iodine. Then titrate the iodine liberated against a standard aqueous solution of sodium thiosulphate.

Stage 1 oxidizing agent being analyzed in sample solution + excess I^- $\longrightarrow$ I_2 + other products

Stage 2 $I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

known volume and concentration



Fig. 55.21 A household bleach containing sodium hypochlorite

Household bleaches usually contain sodium hypochlorite* ($NaOCl$) as the active ingredient. Sodium hypochlorite can oxidize iodide ions to iodine.

$$OCl^-(aq) + 2H^+(aq) + 2I^-(aq) \longrightarrow I_2(aq) + Cl^-(aq) + H_2O(l)$$

We can determine the concentration of sodium hypochlorite in a household bleach by allowing a known volume of the bleach to react with iodide ions to liberate iodine. Then titrate the iodine liberated against a standard aqueous solution of sodium thiosulphate. The titration should be performed rapidly to minimize air oxidation of the iodide ions.

Take the following steps in the experiment:

Step 1	Dilute a 10.0 cm^3 sample of the bleach to 250.0 cm^3 with distilled water.
Step 2	Place 25.0 cm^3 of the diluted bleach in a conical flask. Add excess aqueous solution of potassium iodide and dilute sulphuric acid.
Step 3	(a) Titrate the iodine liberated in the mixture against a standard aqueous solution of sodium thiosulphate until the colour of the mixture changes to pale yellow. (b) At this point, add a few drops of starch solution, producing a dark blue colour. Add the sodium thiosulphate solution dropwise until the mixture becomes colourless.

* Sodium hypochlorite is also called sodium chlorate(I).

ASL 2003 Q12(b)(II)(iii)
ASL 2004 Q8(c)(ii)
AL 2001 Paper 2 Q1(c)(iii)(II)
AL 2009 Paper 1 Q7(a)

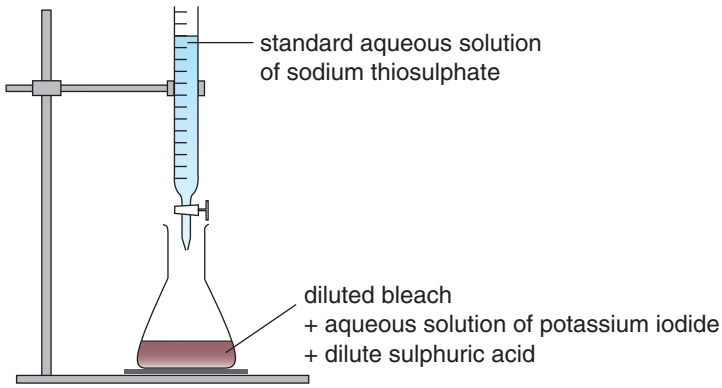
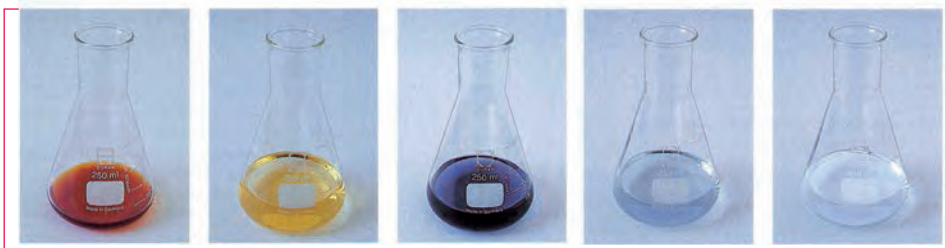


Fig. 55.22 Set-up for titrating the iodine liberated (from the reaction between bleach and potassium iodide) against a standard aqueous solution of sodium thiosulphate

ASL 2012 Paper 1 Q8(a)
AL 2012 Paper 1 Q7(b)(i)



(a) (b) (c) (d) (e)

(a) The mixture is brown in colour due to the presence of iodine
(b) Add a few drops of starch solution when the mixture in the conical flask changes to pale yellow*
(c) A dark blue colour is produced upon the addition of starch solution
(d) The colour intensity of the mixture decreases upon the addition of $Na_2S_2O_3(aq)$
(e) The end point is reached when the mixture becomes colourless

Fig. 55.23 Changes in the reaction mixture during an iodine / thiosulphate titration N4

* As iodine is present in the conical flask throughout the titration process up to the equivalence point, starch solution should not be added until immediately before the equivalence point. Otherwise, some iodine would remain bound to the starch particles even after the equivalence point. We need to add an excess of the titrant before the mixture becomes colourless. This would introduce error in the titration results.

N4 Examination questions often ask about the end point detection during an iodine / thiosulphate titration.

Calculating the concentration of sodium hypochlorite in the bleach

Fig. 55.24 summarizes the steps we can follow to calculate the concentration of sodium hypochlorite in the bleach.

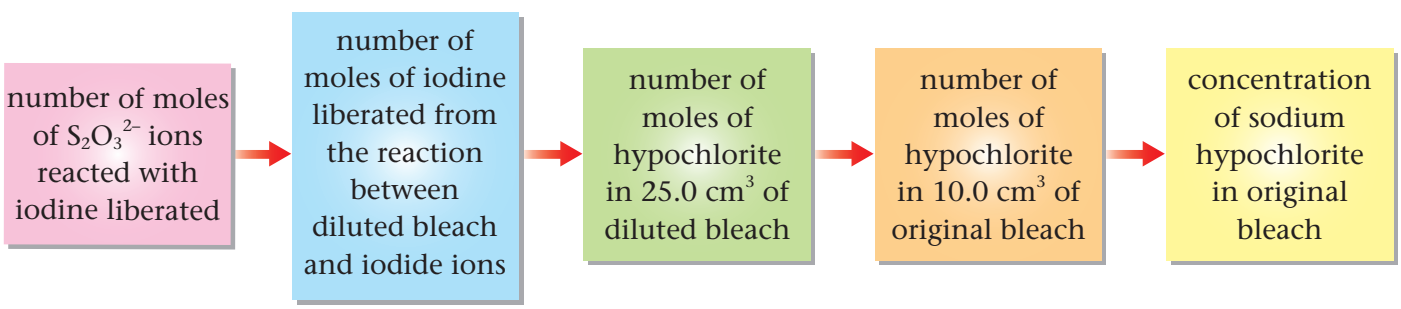


Fig. 55.24 Steps for calculating the concentration of sodium hypochlorite in a bleach

ASL 2003 Q12(b)(II)(ii),(iv)
ASL 2004 Q8(c)(iii)

The following results are obtained in a certain experiment:

Concentration of $Na_2S_2O_3(aq)$ = 0.0500 mol dm^{-3}

Volume of $Na_2S_2O_3(aq)$ required to reach the end point
= 26.4 cm^3

We can calculate the concentration of sodium hypochlorite in the bleach as follows:

$$I_2(aq) + 2S_2O_3^{2-}(aq) \longrightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$
0.0500 mol dm^{-3}
26.4 cm^3

Number of moles of $S_2O_3^{2-}$ ions reacted with the iodine liberated

$$= 0.0500 \text{ mol } dm^{-3} \times \frac{26.4}{1\,000} dm^3$$

$$= 1.32 \times 10^{-3} \text{ mol}$$

AL 2001 Paper 2 Q1(c)(iii)(I)
AL 2006 Paper 2 Q7(c)(ii)

According to the equation, 1 mole of I_2 reacts with 2 moles of $S_2O_3^{2-}$ ions.

i.e. number of moles of I_2 reacted with $S_2O_3^{2-}$ ions

$$= \frac{1.32 \times 10^{-3}}{2} \text{ mol}$$

$$= 6.60 \times 10^{-4} \text{ mol}$$

= number of moles of I_2 liberated from 25.0 cm^3 of diluted bleach

Consider the equation for the reaction between hypochlorite ions and iodide ions in an acidic solution:

$$OCl^-(aq) + 2H^+(aq) + 2I^-(aq) \longrightarrow I_2(aq) + Cl^-(aq) + H_2O(l)$$
25.0 cm^3 diluted bleach — 6.60×10^{-4} mol

According to the equation, 1 mole of OCl^- ions reacts with excess I^- ions to liberate 1 mole of I_2.

i.e. number of moles of OCl^- ions in 25.0 cm^3 diluted bleach
= 6.60×10^{-4} mol

Number of moles of OCl^- ions in 250.0 cm^3 diluted bleach
= $10 \times 6.60 \times 10^{-4}$ mol
= 6.60×10^{-3} mol

Concentration of OCl^- ions in the original bleach

$$= \frac{6.60 \times 10^{-3} \text{ mol}}{\frac{10.0}{1\ 000} \text{ dm}^3}$$

$$= 0.660 \text{ mol dm}^{-3}$$

Therefore the concentration of hypochlorite ions in the bleach is 0.660 mol dm^{-3}.

Suppose the density of the bleach is 1.05 g cm^{-3} and the molar mass of NaOCl is 74.5 g mol^{-1}, we can calculate the percentage by mass of sodium hypochlorite in the bleach as follows:

Mass of 1.00 dm^3 of bleach = 1.05 g cm^{-3} x 1 000 cm^3

= 1 050 g

Mass of NaOCl in 1.00 dm^3 of bleach

= 0.660 mol x 74.5 g mol^{-1}

= 49.2 g

Percentage by mass of NaOCl in the bleach $= \frac{49.2 \text{ g}}{1\ 050 \text{ g}} \times 100\%$

= 4.69%

Therefore the percentage by mass of sodium hypochlorite in the bleach is 4.69%.

In the following activity, you are going to determine the sodium hypochlorite content in a bleach.

Video

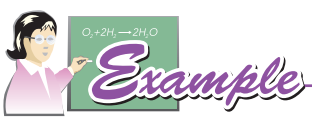
Determining the concentration of sodium hypochlorite in a household bleach using an iodine / thiosulphate titration

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 55.6

Determining the sodium hypochlorite content in a bleach.



ASL 2007 Paper 2 Q3(b)
AL 2007 Paper 2 Q4(c)

55.2 Calculating the percentage by mass of copper in a mineral

The percentage by mass of copper in a 0.200 g sample mineral is determined as follows:

Stage 1	Dissolve the sample in $HNO_3(aq)$ to obtain $Cu(NO_3)_2(aq)$. Add an excess of KI(aq) to the solution. Iodine is liberated.
Stage 2	Titrate the iodine liberated against 0.100 mol dm^{-3} $Na_2S_2O_3(aq)$. The titre is 20.0 cm^3.

a) The reaction in Stage 2 can be represented by the following equation:

$I_2(aq) + 2S_2O_3^{2-}(aq) \longrightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

Calculate the number of moles of iodine liberated from the reaction in Stage 1.

b) The reaction between $Cu(NO_3)_2(aq)$ and KI(aq) in Stage 1 can be represented by the following equation:

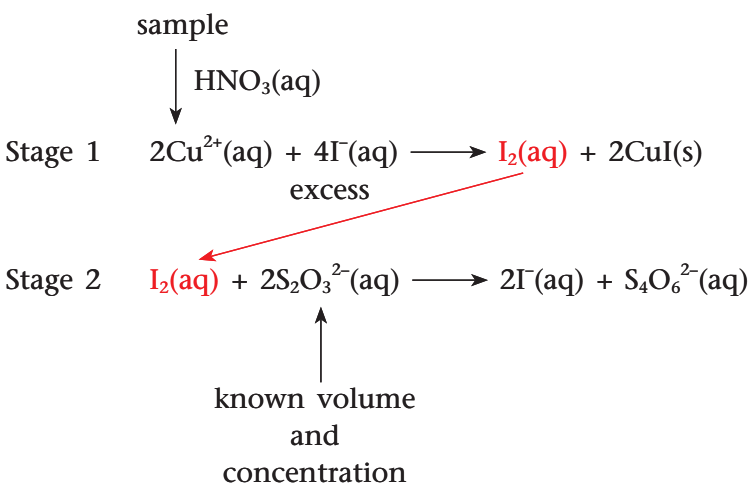
$2Cu^{2+}(aq) + 4I^-(aq) \longrightarrow I_2(aq) + 2CuI(s)$

Calculate the percentage by mass of copper in the sample of mineral.

(Relative atomic mass: Cu = 63.5)

Solution

The figure below summarizes the reactions involved in the experiment.



a)

$I_2(aq)$	+	$2S_2O_3^{2-}(aq)$	$\longrightarrow$	$2I^-(aq)$	+	$S_4O_6^{2-}(aq)$
? mol		0.100 mol dm^{-3}				
		20.0 cm^3				

Number of moles of $S_2O_3^{2-}$ ions reacted with iodine liberated $= 0.100 \text{ mol dm}^{-3} \times \frac{20.0}{1\ 000} \text{ dm}^3$

$= 0.00200 \text{ mol}$

According to the equation, 1 mole of I_2 reacts with 2 moles of $S_2O_3^{2-}$ ions.

i.e number of moles of I_2 reacted with $S_2O_3^{2-}$ ions

$= \frac{0.00200}{2}$ mol

= 0.00100 mol

= number of moles of I_2 liberated in the reaction in Stage 1

b)

$$2Cu^{2+}(aq) + 4I^-(aq) \longrightarrow I_2(aq) + 2CuI(s)$$
$$\text{? mol} \qquad\qquad 0.00100 \text{ mol}$$

According to the equation, 2 moles of Cu^{2+} ions react to give 1 mole of I_2.

i.e number of moles of Cu in the sample = 2 x 0.00100 mol

= 0.00200 mol

Mass of Cu in the sample = 0.00200 mol x 63.5 g mol^{-1}

= 0.127 g

Percentage by mass of Cu in the sample = $\frac{0.127 \text{ g}}{0.200 \text{ g}}$ x 100%

= 63.5%

∴ the percentage by mass of copper in the sample is 63.5%.

Checkpoint

Refer to page T62.

1 Sulphur dioxide is used to preserve wines. In an experiment to determine the sulphur dioxide content of a white wine, 25.0 cm^3 of the sample were titrated against a 0.00450 mol dm^{-3} iodine solution quickly. 15.8 cm^3 of the iodine solution were required to reach the end point. N5

The reaction involved can be represented by the following equation:

$$SO_2(g) + I_2(aq) + 2H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2I^-(aq) + 4H^+(aq)$$

a) Why should the titration be carried out quickly?

b) Briefly describe how the end point of the titration could be determined.

c) Calculate the concentration of sulphur dioxide, in mg dm^{-3}, in the sample of white wine.
182 mg dm^{-3}

(Relative atomic masses: O = 16.0, S = 32.1)

DSE 2013 Paper 2 Q3(b)(ii),(iii),(iv)

N5 This method cannot be used to determine the sulphur dioxide of a red wine. The intense red colour of red wine may mask the colour of the dark blue complex, making it difficult to detect the end point.

2 The active ingredient in a bleaching powder is calcium hypochlorite ($Ca(OCl)_2$). When reacting with a dilute acid, calcium hypochlorite liberates chlorine readily.

$$Ca(OCl)_2(s) + 4HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l) + 2Cl_2(g)$$

The chlorine liberated is essential for the bleaching process and is known as 'available chlorine'. The chlorine liberated can react with potassium iodide to form iodine.

The following experiment is carried out to determine the percentage of 'available chlorine' in a sample of bleaching powder.

Stage 1	Add a 0.285 g sample of bleaching powder to 100 cm^3 of water. Treat with an excess of KI(aq) and HCl(aq) to form iodine.
Stage 2	Titrate the iodine formed against 0.100 mol dm^{-3} $Na_2S_2O_3(aq)$. 24.6 cm^3 of the thiosulphate solution are required to reach the end point.

a) i) Write an equation for the reaction that occurs in Stage 2.

ii) Calculate the number of moles of iodine formed in Stage 1. 1.23×10^{-3} mol

b) The reaction between chlorine from the bleaching powder and KI(aq) can be represented by the following equation:

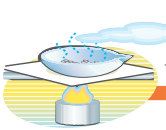
$$Cl_2(aq) + 2I^-(aq) \longrightarrow I_2(aq) + 2Cl^-(aq)$$

Calculate

i) the mass of 'available chlorine' in the bleaching powder; 8.73×10^{-2} g

ii) the percentage by mass of 'available chlorine' in the bleaching powder. 30.6%

(Relative atomic mass: Cl = 35.5)



Summary

1 a) Two common quantitative methods of analysis are gravimetric analysis and volumetric analysis.

b) Gravimetric analysis involves determining the mass of the species being analyzed or a compound chemically related to it.

Extension

c) Volumetric analysis involves measuring the volume of a solution containing sufficient reagent to react completely with the species being analyzed in a process called titration.

Extension

2 An experimental error can be classified as either systematic or random.

3 Examples of gravimetric analysis:

a) determining the phosphorus content in a sample of fertilizer by precipitating the phosphorus as magnesium ammonium phosphate ($MgNH_4PO_4 \bullet 6H_2O$);

b) determining the calcium content in a sample solution by precipitating the calcium as calcium oxalate ($CaC_2O_4 \bullet H_2O$).

4 The concentration of chloride ions in a sample solution can be determined by precipitation titration with a standard aqueous solution of silver nitrate.

5 Examples of redox titrations involving an aqueous solution of potassium permanganate:

a) determining the iron content in commercial iron tablets;

b) analyzing the quality of water by determining its permanganate index.

6 Redox titrations involving iodine:

a) determining the concentration of a reducing agent in a sample solution by measuring the amount of iodine required to undergo a redox reaction with the reducing agent in a known volume of the sample.

$$\text{reducing agent} + I_2 \longrightarrow I^- + \text{other products}$$

b) determining the concentration of an oxidizing agent by allowing the oxidizing agent in a known volume of the sample solution to react with excess iodide ions to liberate iodine. Then titrate the iodine liberated against a standard aqueous solution of sodium thiosulphate.

Stage 1 oxidizing agent being analyzed in sample solution + excess I^- $\longrightarrow$ I_2 + other products

Stage 2 $I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

known volume and concentration

Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

Extension

1 a) Explain the meaning of the term 'gravimetric analysis'.
Refer to page T63.
b) Give TWO examples of gravimetric analysis.

2 List FIVE factors you would consider when deciding on an analytical method.
Refer to page T63.

Extension

3 Give one example for each type of systematic error.

a) Instrumental error
A pipette / burette may hold / deliver a volume slightly different from the volume indicated by the manufacturer.
b) Operative error
During a titration, a student who is insensitive to colour changes may use more reagent.
c) Error of method
During a titration, a small excess of the reagent is added to cause an indicator to change colour.

4 a) Explain why an aqueous solution of potassium permanganate requires standardization before its use for titration.
Refer to page T63.
b) Briefly describe how you can standardize an aqueous solution of potassium permanganate.

5 a) Explain why standard iodine solutions are NOT prepared directly from solid iodine.
Refer to page T64.
b) A standard iodine solution is prepared by dissolving 1.07 g of $KIO_3(s)$ in excess KI(aq) and $H_2SO_4(aq)$, and then making up the solution to 250.0 cm^3.

i) Write an ionic equation for the reaction taking place.

ii) Calculate the molarity of the standard iodine solution. 6.00×10^{-2} mol dm^{-3}

(Relative atomic masses: O = 16.0, K = 39.1, I = 126.9)

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 0.250 g of an impure sample of potassium chloride was dissolved in water. The solution was treated with excess aqueous solution of silver nitrate. 0.291 g of silver chloride was obtained. Calculate the percentage by mass of potassium chloride in the sample.
60.4%; refer to page T64.
(Relative atomic masses: Cl = 35.5, K = 39.1, Ag = 107.9)

7 Tin forms an oxide, A, that contains the metal in both oxidation states II and IV. The formula of A can be found by the following method.
Refer to page T64.

Step 1 A sample of A was dissolved in $H_2SO_4(aq)$, producing solution B, which was a mixture of tin(II) sulphate and tin(IV) sulphate.

Step 2 25.0 cm^3 of solution B were titrated against 0.0150 mol dm^{-3} $KMnO_4(aq)$. 18.00 cm^3 of $KMnO_4(aq)$ were required to reach the titration end point.

Step 3 Another 25.0 cm^3 sample of solution B was treated with an excess of powdered zinc. This converted all the tin into tin(II) ions. The excess of zinc powder was filtered off. The filtrate was titrated against 0.0150 mol dm^{-3} $KMnO_4(aq)$. 27.15 cm^3 of $KMnO_4(aq)$ were required to reach the titration end point.

The equation for the reaction occurring during the titration is as follows.

$$2MnO_4^-(aq) + 16H^+(aq) + 5Sn^{2+}(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5Sn^{4+}(aq)$$

a) Write an ionic equation for the reaction between Zn(s) and $Sn^{4+}(aq)$.

b) Use the results of the two titrations to calculate

i) the number of moles of Sn^{2+} in the titration sample in *Step 2*; 6.75×10^{-4} mol

ii) the number of moles of Sn^{2+} in the titration sample in *Step 3*. 1.02×10^{-3} mol

c) Use the results of your calculations in (b) to deduce the ratio of Sn^{2+} / Sn^{4+} in the oxide A, and hence suggest the chemical formula of A. Sn_3O_4

8 A 25.0 cm^3 sample of sea water was diluted to 250.0 cm^3. 25.0 cm^3 portions of the diluted sea water was titrated against 0.0750 mol dm^{-3} aqueous solution of silver nitrate. The mean titre was 19.8 cm^3.
Refer to page T65.

a) i) Suggest a suitable indicator for the titration.

ii) State and explain the colour change at the end point.

b) Calculate the concentration of chloride ions in the sea water sample in g dm^{-3}. 21.2 g dm^{-3}

(Relative atomic mass: Cl = 35.5)

9 Hydrogen peroxide (H_2O_2) is a mild oxidizing agent that is used in the restoration of oil paintings. Paintings darken over time as some of the metal ions in the paints react with atmospheric pollutants. Hydrogen peroxide can be used to convert these unwanted dark coloured compounds to white products.

Refer to page T65.

a) Write down the ionic half-equation for hydrogen peroxide, H_2O_2, acting as an oxidizing agent. This reaction takes place in acid solution and water is the only product.

b) The concentration of an H_2O_2 solution can be found by titration of samples of the solution with an aqueous solution of potassium permanganate of known concentration using acidic conditions.

i) Describe how this titration would be carried out. Give clear experimental details and state how the end point is determined.

ii) A solution of hydrogen peroxide is diluted by a factor of 10.0.

10.0 cm^3 of this diluted H_2O_2 reacted with exactly 18.2 cm^3 of 0.0200 mol dm^{-3} MnO_4^- solution.

The equation for the reaction taking place is given below.

$$2MnO_4^-(aq) + 6H^+(aq) + 5H_2O_2(aq) \longrightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5O_2(g)$$

Calculate the concentration of the undiluted H_2O_2 solution. 0.910 mol dm^{-3}

iii) The concentration of the hydrogen peroxide solution used for treating paintings must not be greater than 3.0%. Assume this means 3.0 g of H_2O_2 in 100 cm^3 of solution.

Is the undiluted solution of H_2O_2 suitable to be used for treating paintings?

(Relative atomic masses: H = 1.0, O = 16.0)

(OCR Advanced GCE (Salters), Chemistry of Materials, Jun. 2006)

10 A student carried out an experiment to determine the percentage by mass of iron in an iron wire. A sample of 0.212 g of the iron wire was added to 100 cm^3 of 2 M sulphuric acid. The solution obtained was then titrated against 0.0208 M potassium permanganate solution. 34.70 cm^3 of the potassium permanganate solution was required to reach the end point.

a) Write the ionic equation for the reaction taking place during the titration.

b) Calculate the percentage by mass of iron in the iron wire sample.

c) State one assumption for the experiment.

d) Could hydrochloric acid be used instead of sulphuric acid in this experiment? Explain your answer.

(HKASLE 2006)

11 An experiment, consisting of the following two stages, was carried out to determine the content of ascorbic acid in a sample of grapefruit juice.

Stage 1: Excess KI(aq) and H_2SO_4(aq) were added to 100.0 cm^3 of 0.0240 M KIO_3(aq). The mixture was then diluted to 250.0 cm^3 using deionized water to give a standard iodine solution.

Stage 2: 25.0 cm^3 portions of the grapefruit juice were titrated against the standard iodine solution. The mean titre was 18.0 cm^3.

a) The reaction in Stage 1 can be represented by the following equation:

$$IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \longrightarrow 3I_2(aq) + 3H_2O(l)$$

Calculate the molarity of the standard iodine solution prepared.

b) Suggest how the end point of the titration in Stage 2 can be detected.

c) The reaction of ascorbic acid with iodine can be represented by the following equation:

$$\underset{\text{ascorbic acid}}{C_6H_8O_6(aq)} + I_2(aq) \longrightarrow \underset{\text{dehydroascorbic acid}}{C_6H_6O_6(aq)} + 2H^+(aq) + 2I^-(aq)$$

Calculate the mass of ascorbic acid in 1.0 dm^3 of the sample of grapefruit juice.

(HKASLE 2003)

12 The element iodine can be produced from the mineral Chile saltpeter, which contains sodium iodate, $NaIO_3$. The iodate ions are converted to iodine in a two-step process.

Refer to page T66.

Ionic equations for the reactions are shown below.

Step 1 $IO_3^-(aq) + 3HSO_3^-(aq) \longrightarrow I^-(aq) + 3HSO_4^-(aq)$

Step 2 $IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \longrightarrow 3I_2(aq) + 3H_2O(l)$

a) i) Describe a test you could carry out to confirm the presence of iodide ions in a solution. Indicate the result of the test.

ii) Identify the TWO elements in *Step 1* which show a change of oxidation number during the reaction. Give their initial and final oxidation numbers.

iii) The equation for *Step 1* shows that 1 mole of iodate ions, IO_3^-, reacts with 3 moles of hydrogensulphite ions, HSO_3^-. Show that your answer to (ii) agree with this ratio.

b) An experiment was carried out to determine the concentration of the iodine prepared in *Step 2*.

A 10.0 cm^3 portion of the iodine solution was titrated with sodium thiosulphate solution of concentration 0.0100 mol dm^{-3}. The volume of sodium thiosulphate solution added at the end point was 24.0 cm^3.

The equation for the reaction is

$I_2(aq) + 2Na_2S_2O_3(aq) \longrightarrow 2NaI(aq) + Na_2S_4O_6(aq)$

i) What piece of apparatus would you use to measure out the 10.0 cm^3 portion?

ii) Suggest a suitable indicator to show the end point of this titration. State the colour change you would see.

iii) Calculate the number of moles of sodium thiosulphate used in the titration.
2.40×10^{-4} mol

iv) Calculate the number of moles of iodine which reacted with the sodium thiosulphate solution.
1.20×10^{-4} mol

v) Calculate the concentration, in mol dm^{-3}, of the iodine solution.
0.0120 mol dm^{-3}

(Edexcel Advanced Subsidiary GCE (Nuffield), Unit Test 2, Jun. 2006)

13 Household bleach contains NaOCl as the active ingredient. In an experiment to determine the concentration of NaOCl in a brand of household bleach, a sample of the bleach was diluted 50 times using deionized water. To 25.00 cm^3 of the *diluted* bleach, excess KI(aq) and $H_2SO_4(aq)$ were added. The mixture was then titrated against 0.0552 M $Na_2S_2O_3(aq)$. 16.00 cm^3 of the $Na_2S_2O_3(aq)$ was required to reach the end point.

a) Describe how the sample of bleach can be diluted using apparatus available in a school laboratory.

b) Suggest how the end point of the titration can be detected.

c) The reaction of $OCl^-(aq)$ with $I^-(aq)$ under acidic conditions can be represented by the equation:

$OCl^-(aq) + 2I^-(aq) + 2H^+(aq) \longrightarrow Cl^-(aq) + I_2(aq) + H_2O(l)$

From the titration result, calculate the concentration of $OCl^-(aq)$, in mol dm^{-3}, in this brand of bleach.

(HKASLE 2004)

14 In an experiment to determine the percentage by mass of copper in a sample of copper ore, 3.00 g of the sample was warmed with excess $HNO_3(aq)$ to convert all copper-containing substances to $Cu^{2+}(aq)$ ions. The resulting solution was boiled to remove all nitrogen oxides formed. Upon cooling, the solution was diluted to 250.0 cm^3. 25.00 cm^3 of the diluted solution was withdrawn and then treated with excess KI(aq) to liberate CuI(s) and $I_2(aq)$. The titration of the $I_2(aq)$ liberated required 25.80 cm^3 of 0.102 M $Na_2S_2O_3(aq)$.

a) Write chemical equations for

i) the reaction of $Cu^{2+}(aq)$ with $I^-(aq)$, and

ii) the reaction of $I_2(aq)$ with $Na_2S_2O_3(aq)$.

b) Calculate the percentage by mass of copper in the sample of copper ore.

c) State ONE assumption made in the experiment.

(HKALE 2007)

Exercise

15 You are provided with the following materials:

- water sample
- 2 mol dm^{-3} sulphuric acid
- 0.00200 mol dm^{-3} potassium permanganate solution
- 0.00500 mol dm^{-3} sodium oxalate solution

Design an experiment to determine the permanganate index of the water sample.
(You may use apparatus commonly available in a school laboratory.)

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

Place 25.0 cm^3 of the water sample in a conical flask using a pipette.
Add 5 cm^3 of dilute sulphuric acid and mix.
Place the conical flask in a boiling water bath for 10 minutes.
Add 5.00 cm^3 of the aqueous solution of potassium permanganate to the water sample using a pipette.
After 10 minutes, add 5.00 cm^3 of the aqueous solution of sodium oxalate using a pipette. Wait until the solution becomes colourless.
Fill a burette with the aqueous solution of potassium permanganate. Run the aqueous solution of potassium permanganate into the hot mixture until the first appearance of a persistent pale pink colour.
Initially, 5.00 cm^3 of $KMnO_4(aq)$ and 5.00 cm^3 of $Na_2C_2O_4(aq)$ are added to the water sample.

Number of moles of $KMnO_4$ in 5.00 cm^3 solution $= 0.00200 \text{ mol } dm^{-3} \times \frac{5.00}{1\,000} dm^3$
$= 1.00 \times 10^{-5}$ mol

Number of moles of $Na_2C_2O_4$ in 5.00 cm^3 solution $= 0.00500 \text{ mol } dm^{-3} \times \frac{5.00}{1\,000} dm^3$
$= 2.50 \times 10^{-5}$ mol

Under acidic conditions, the oxalate ions are converted to oxalic acid. Potassium permanganate reacts with the oxalic acid according to the following equation:
$5H_2C_2O_4(aq) + 2MnO_4^-(aq) + 6H^+(aq) \longrightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$
According to the equation, 5 moles of $C_2O_4^{2-}$ ions react with 2 moles of MnO_4^- ions.
If none of the permanganate ion is consumed by the oxidizable matter in the water sample, all the oxalate ions will react with the permanganate ions and no oxalate ion will remain in the mixture. Suppose V cm^3 of aqueous solution of permanganate are required to react with the oxalate ions remaining in the mixture. It can be deduced that this amount of permanganate ions is consumed in oxidizing the oxidizable matter in the water sample.
Number of moles of MnO_4^- ions consumed in oxidizing matter in water sample $= 0.00200 \text{ mol } dm^{-3} \times \frac{V}{1\,000} dm^3$
$= 2V \times 10^{-6}$ mol
$= 2V \times 10^{-3}$ mmol
Number of millimoles of O_2 required to oxidize the same amount of matter $= 1.2 \times 2V \times 10^{-3}$ mmol
Mass of O_2 required to oxidize the same amount of matter $= 2.4V \times 10^{-3} \text{ mmol} \times 32.0 \text{ mg mmol}^{-1}$
$= 7.68V \times 10^{-2}$ mg

Permanganate index of water sample $= \frac{7.68V \times 10^{-2} \text{ mg}}{\frac{25.0}{1\,000} \text{ L}}$
$= 3.07V$ mg of O_2 per litre

Exercise